Exp-4 Setting Up a Local Area Network (LAN)

Aim:-
To design, set up and configure a Local Area Network (LAN) connecting four computers through a network switch, assign static IP Addresses to all devices, verify connectivity using the ping command, and demonstrate resources / data sharing across the network.

IP Address Configuration Table:-

Device	IP Address	Subnet Mask	Connected To
PC0	10.1.1.1	255.0.0.0	Switch Port Fa0/1
PC1	10.1.1.2	255.0.0.0	Switch Port Fa0/2
PC2	10.1.1.3	255.0.0.0	Switch Port Fa0/3
PC3	10.1.1.4	255.0.0.0	Switch Port Fa0/4
Switch (Management)	10.1.1.5	255.0.0.0	VLAN 1 Interface

Implementation:

1) **Place the Devices**

* Open Cisco Packet Tracer.
* From the device category panel, select End Devices and drag 4 PC's onto the Workspace
* select Switches and drag 1 switch (2960 series) onto the Workspace.

ii) Cable the Devices

* Click Connections → select Copper straight-Through cable.
* Click each PC, then click the switch and select an available Fast Ethernet port to complete each Connection.
* Repeat for all four PC's (port Fa 0/1 to Fa 0/4). Green link lights confirm a successful physical connection.

iii) Assign Ip Address to PC's

* click a PC → Desktop tab → Ip Configuration
* Enter the static Ip Address and subnet mask as per the Ip table for each PC.

iv) Configure the switch Mangement Ip (optional)

* click the switch → CLI tab
* Enter the following commands to assign a Management IP.

```
enable
configure terminal
interface vlan 1
ip address 10.1.1.5 255.0.0.0
no shutdown
exit
```

Scanned with OKEN Scanner

v) Verify Connectivity Using Ping

* click a PC → Desktop tab → Command prompt
* Use the ping command to test Connectivity to the other PC's . eg 10.1.1.2
* Successful "Reply from ...." message confirm connectivity.

vi) Demonstrate Data sharing (Server Simulation)

* Add a Server device (End device → Server-PT) and Connect it to the switch.
* Assign it an IP address (eg. 10.1.1.10 subnetmask 255.0.0.0) via its desktop → IP Configuration tab.
* Open the server's services (config) tab and enable the HTTP Service
* From another PC, open the Web browser and enter the server's IP address to load the page, confirming successful data transfer across the LAN.

Result:-

Thus to design, set up and configure a LAN connecting four computers through a network switch and assign IP address and verify the Connectivity is successfully demonstrated.

Scanned with OKEN Scanner

Server-PT
Server0
PC-PT
PC0
PC-PT
PC1
PC-PT
PC2
2960-24TT
Switch0
PC-PT
PC3
PC0
Physical
Config
Desktop
Programming
Attributes
Web Browser
URL
http://10.0.0.10
Go
Stop
Cisco Packet Tracer
Welcome to Cisco Packet Tracer. Opening doors to new opportunities. Mind Wide Open.
Quick Links:
A small tour
Copyright
Privacy Policy
Images
Realtime
Simulation
Time: 00:05:14
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Delete